higlass-python

A Programmable Genome Browser for Linked Interactive Visualization and Exploration of Genomic Data

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pip install higlass-python
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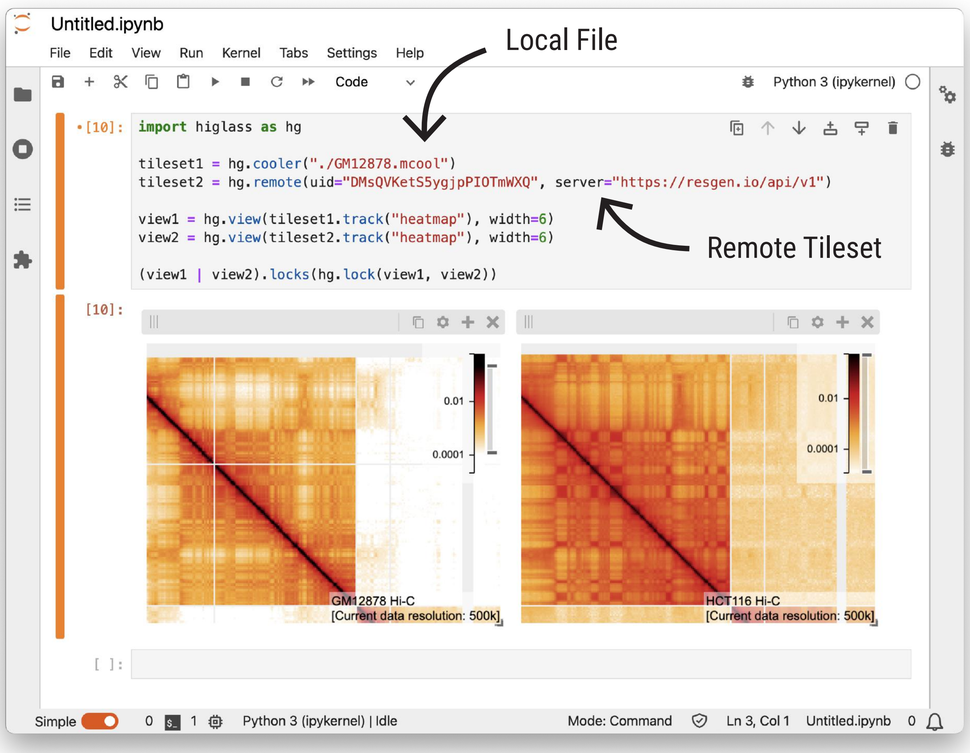
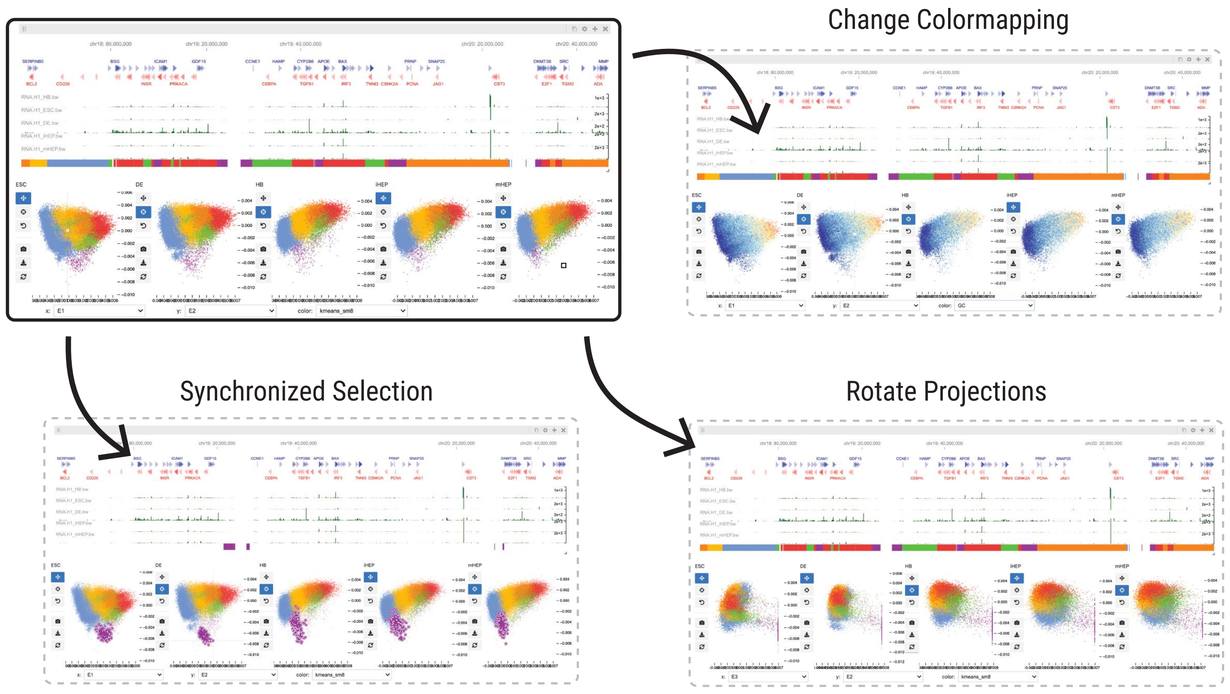
Scriptable Genome Browser



Genomic Loci Embedding Explorer

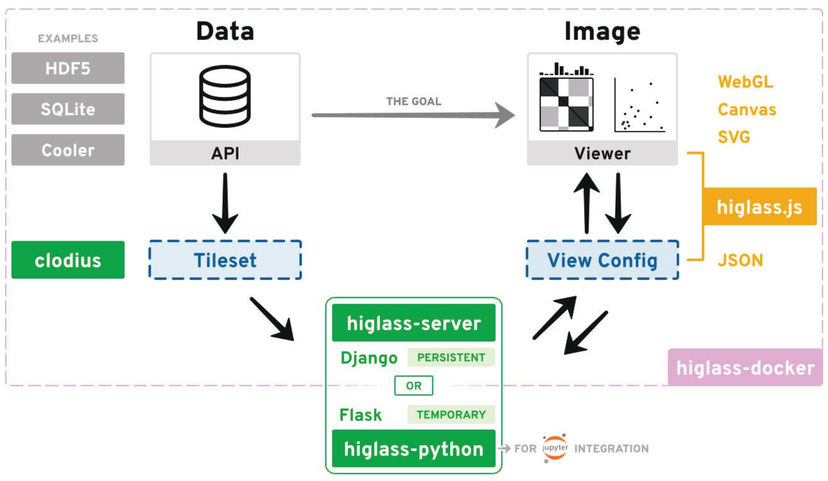


higlass-python is a Python-based software development kit (SDK) that enables the creation of custom genome browser applications and flexible exploration of genomic data. It embeds visualizations directly into computational notebooks, enabling computational biologists to tailor their analytical environments with resources from the scientific Python ecosystem.

We built our SDK around HiGlass, a web-based genome application designed for the visualization of 1D and 2D genomic datasets (Kerpedjiev et al 2018). It provides smooth navigation via continuous zoom and pan interactions and includes flexible view configuration.

The tight integration with Python and Jupyter allows bidirectional linking of genome browser state (e.g., genomic coordinates) with external notebook components (scatter plots, statistical plots, drop down menus, etc.). Third-party visualizations can access selections of genomic ranges within a HiGlass view and exercise control over the mapping of individual selections back to HiGlass.

Architecture



Tileset: A server-based abstraction for efficient multi-resolution genomic data provision to the viewer.
View Config: A JSON specification that describes the viewer configuration and arrangement.
Track: Visualization implementation in the viewer. Novel track types can be registered as plugins.

We employed our toolkit to examine the latent embedding spaces of 3D genomic contact frequency profiles within a functional and epigenomic context. The representations of contact profiles are embedded into 2D visualizations where each point represents a genomic locus or interval. Our SDK coordinates multiple views of individual loci with HiGlass, demonstrating how individual selections of loci are synchronized across all views. A key feature of this use case is the ability to load, explore, export, and analyze the data all within the same computational environment.

higlass-python offers a simplified Python API to:

Compose tilesets, tracks, views, and synchronizations

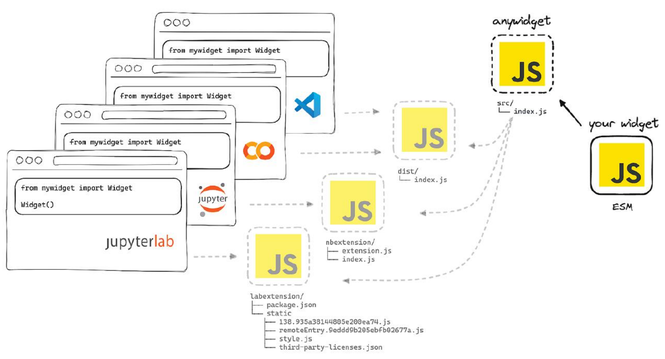
Load, edit, save and validate HiGlass view configurations

Federate data from local and remote data sources

Embed interactive visualizations within Jupyter Notebooks, JupyterLab, VSCode, and Google Colab

Core Technology

Jupyter Widgets enrich notebooks with interactive JavaScript-based views and controls for Python objects in the Jupyter kernel. However, the creation of custom widgets has traditionally presented a steep learning curve, requiring an in-depth understanding of both Python and JavaScript.

We introduce **anywidget**, a tool designed to drastically simplify Jupyter widgets for both domain scientists and seasoned developers. It connects interactive visualizations that interface with HiGlass via our SDK, enabling diverse data representations for comprehensive genomic exploration. Although **anywidget**'s capabilities extend beyond genomic visualizations, it proves highly useful in this context.

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pip install anywidget
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